

Author: C. Tarkington (Pellissippi State Community College, Knoxville, TN 37932)

Mentor: L.A Anderson, Geospatial Science and Human Security (Oak Ridge National Laboratory, Oak Ridge, TN 37830)

Introduction

Ring LLC is an Amazon-owned security and smart home system manufacturer. They provide a respectable selection of smart doorbells, home security cameras, and alarm systems. Following comparison analysis against reputable firms like ADT and SimpliSafe, Ring was named the Best Security Cameras and Video Doorbells. Given how these devices are being installed in homes worldwide, it is imperative that their security features perform well. The main objective of this study was to find potential weaknesses that hostile actors could exploit using network side attack approaches. Amazon and Ring, in turn, have taken the appropriate steps to guarantee that their devices and transmitted data are secure and encrypted against these common attack entries.

Equipment

- Ring Doorbell Camera V.2
- ThinkPad V.7
- Kali Linux



<https://www.androidheadlines.com/2020/05/ring-gen-doorbell-improvements-solar-charging-video-specs-design.html>

Methods and Results

1. Wire Shark: Packet Sniffing

- Captured packets of the Ring camera over the network. Pulled MAC address of the Ring device and the MAC of the access point (AP) it is connected to. Grabbed QOS Data packets but completely encrypted.

```
Type/Subtype: QoS Data (0x0028)
Frame Control Field: 0x8841
.000 0000 0011 0000 = Duration: 48 microseconds
Receiver address: 96:83:c4:11:ad:4d (96:83:c4:11:ad:4d)
Transmitter address: Ring_c3:62:f7 (64:9a:63:c3:62:f7)
Destination address: GLTechnologi_11:ad:4c (94:83:c4:11:ad:4c)
Source address: Ring_c3:62:f7 (64:9a:63:c3:62:f7)
BSS Id: 96:83:c4:11:ad:4d (96:83:c4:11:ad:4d)
STA address: Ring_c3:62:f7 (64:9a:63:c3:62:f7)
.... .... 0000 = Fragment number: 0
0001 1110 0010 .... = Sequence number: 482
Frame check sequence: 0x5355c80d [correct]
[FCS Status: Good]
[WLAN Flags: .p.....TC]
Qos Control: 0x0000
CCMP parameters
Data (1268 bytes)
Data [truncated]: 6a87a98ecc35b54b48ff4ada7865138e64979743
[Length: 1268]
```

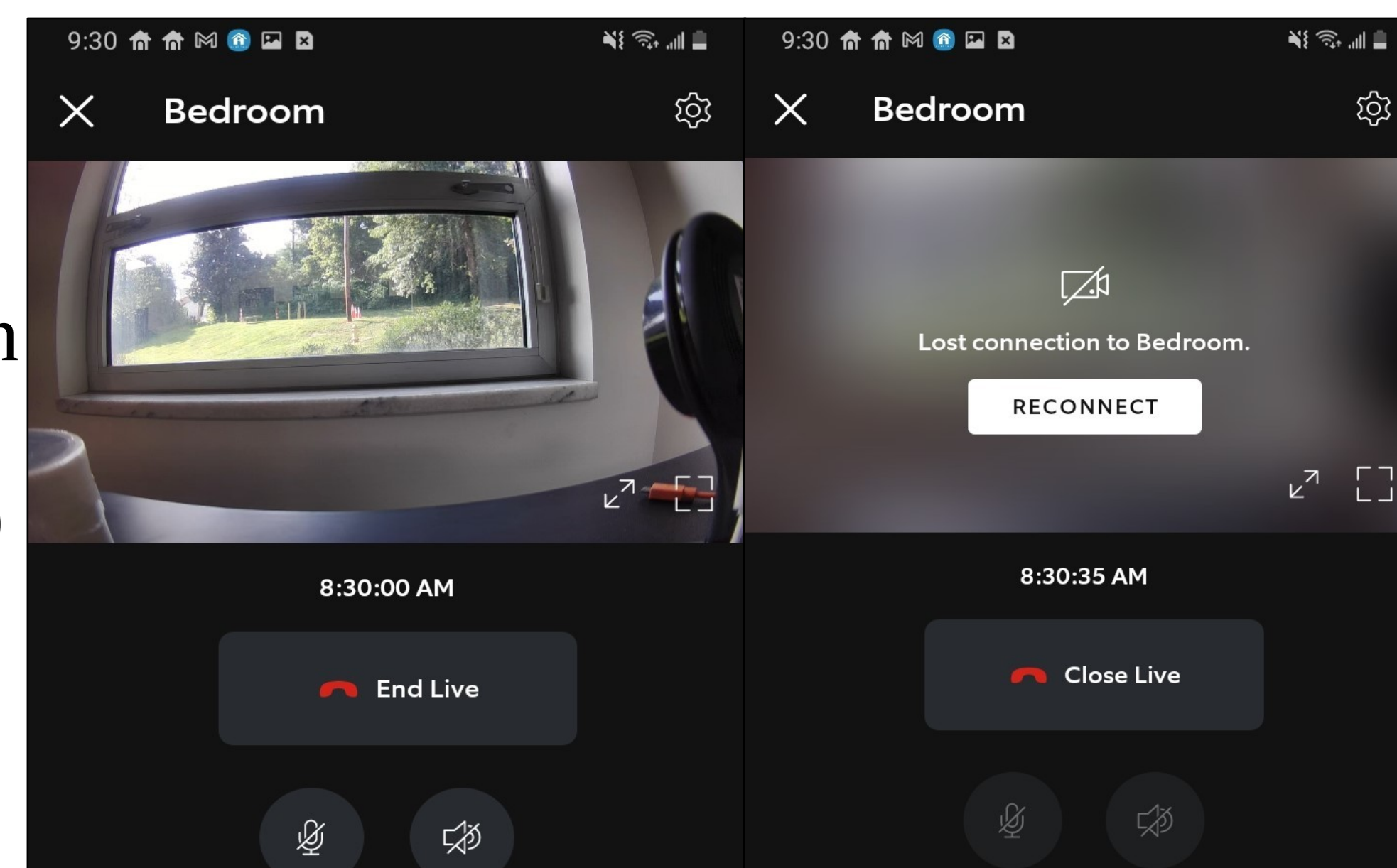
```
(iotuser2@IoT-Kali-1)-[~]
$ sudo nmap -n -Pn -sT -p- 192.168.8.199

Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-07-09 13:09 EDT
Nmap scan report for 192.168.8.199
Host is up (0.094s latency).
All 65535 scanned ports on 192.168.8.199 are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 44.47 seconds
```

2. Nmap: Port Scanning

- Using Nmap to scan any open ports on the device shows that all ports are in ignored states. This typically means the device refused connection. Could be that Ring has an always on encryption mechanism/firewall that rejects the scan.



Discussion

Ring states their devices communicate over the following ports: HTTP (port 80), HTTPS (port 443), DNS (port 53), NTP (port 123), and TCP (port 8557). The socket connection to their cloud servers on ports 443/9002 is used to set up Live View Video. In addition to their standard (in transit) and (at rest) video encryption, Ring introduced Video E2EE (End to End Encryption). With Video E2EE, only the specific enrolled mobile device has the special key needed to unlock these videos, designed so no one else can view the videos -- not even Ring or Amazon. Current devices are not compatible with WPA3 encryption but use WPA2 and Z-Wave protocol encryption.

Future Research

Ring's website offers their source code for download. Further research will be conducted to examine this code and search for any potential weaknesses or techniques that could bypass their encryption mechanisms. The infrastructure of the devices also pose as another entry point. Firmware extraction methods using a JTAGulator will be used to see if there are any opportunities to get around software-based security measures or obtain remote access over the device.

Acknowledgements

This work was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS) under the Science Undergraduate Laboratory Internships Program